

Experiments in Vector Database Ingestion and Retrieval

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Introduction

In this document, I summarise results from multiple experiments I conducted to optimise the RAG pipeline for **AI Market Intelligence Analyst (AMIA)** - an Agentic AI powered Market Analysis tool I created.

Executive Summary

Goal: The goal of the experiments was to find optimal techniques to ingest and retrieve data from a vector database to improve accuracy of the RAG pipeline for AMIA.

Metrics: The primary metric used to evaluate accuracy was whether the pipeline was able to identify relevant and specific information matching the user query versus returning information from generic boilerplate statements that reflected token-match bias.

Result: Best results were obtained with Hierarchical (Parent-Child) chunking and metadata based ingestion and query. Metadata was based on SWOT themes derived from parent chunks (during ingestion) and from user query (during lookup). Token match bias was minimized through metadata matching and semantic content was much improved through the additional context provided by parent chunks.

Experimental Setup and Parameters

Following are the setup and parameters used during the experiments.

- **LLM:** Ollama (Llama3.1)
- **Vector database:** Chroma DB <version>. The documents ingested into the vector database were:
 1. 10-K filing report from NVDA for 2025 and 2026
 2. 10-Q filing report from NVDA for Q1 2027
 3. ArcInvest Big Ideas 2026
 4. Responsible AI – Overcoming barriers and Risks (2026) from McKinsey
- **Ingestion Strategy:**
 1. Flat chunking with default chunk size with overlap,
 2. Hierarchical chunking with Metadata: Parent and child chunks with overlap. Child chunks added to vector DB, parent chunks added to Child chunks as metadata. The LLM was used to derive SWOT themes from the parent chunks. These themes were added to child chunks during ingestion.
- **Embedding Model:** nomic-embed-text, bge-m3
- **Retrieval Strategy:** Plain user query based, Hypothetical Embeddings (HyDELite), Agent Filters, Query Prefixes, Cross Encoders, User query augmented with theme based metadata

- **Retrieval Variables:** 5 and 3 without ranking, 6, 10 or 30 with ranking

All experiments were run on my Apple laptop (M4, 16 GB)

Results Matrix

Following table summarises the results from the experiments.

Query Used: What are the core market growth risks and challenges for NVDA in calendar 2026?

Exp #	Ingestion Variables	Embedding Model	Retrieval Variables	Key Findings
1	Flat Chunking Chunk Size=600 Chunk Overlap=100	nomic_embed_text	k=5 User Query based lookup	Keyword matching overload is obvious from the output. Boilerplate statements from the reports dominate.
2	HyDELite with Agent Filter Chunk Size=600 Chunk Overlap=100	nomic_embed_text	k=6->3	The semantic content improved but some boilerplate statements continue to show up in the output.
3	Query Prefixes search_query: and search:database keywords used for lookup and ingestion Chunk Size=600 Chunk Overlap=100	nomic_embed_text	k=3	No improvement over Flat Chunking (Exp #1). Keyword matching overload is obvious from the output. Boilerplate statements from the reports dominate.
4	Cross Encoders Chunk Size=600 Chunk Overlap=100	nomic_embed_text Cross-Encoder model: ms-marco-MiniLM-L-6-v2	k=30->3	No significant improvement over Flat Chunking (Exp #1). Keyword matching overload is obvious from the output. Boilerplate statements from the reports dominate.
5	HyDELite with Agent Filter Chunk Size=600 Chunk Overlap=100	bge-m3	k=10->5	Similar quality of output as Exp #2. Changing encoding to bge-m3 did not make a significant difference.
6	Hierarchical Chunking with Metadata Child Chunk: Size=400 Overlap=50 Parent Chunk: Size=2000 Overlap=200 SWOT Themes based metadata used for ingestion and query	bge-m3	k=3	Token match bias reduced dramatically. Semantic content improved significantly. No boilerplate statements from standard disclaimers in 10-K and 10-Q statements. Specific and accurate content in the final response.

Analysis and Summary

Following observations were made based on the results listed in the table above.

- **Experiment #1 vs #3: Key variable change is addition of Query Prefixes.** This change is specific to the `nomic_embed_text` model. Prefixes are added to the query (`search_query:`) and to the chunks stored in the vector database during ingestion (`search_document:`). This tells the embedding model what specific task you are performing so it can generate accurate, context-aware vector representations. This is known to improve semantic matching for the `nomic_embed_text` embedding model.
This did not yield any improvements in my experiments.
- **Experiments #2 vs #5: Key variable change is the embedding model, from `nomic_embed_text` to `bge-m3`.**
`nomic-embed-text` is a lightweight model that uses 768-dimension vectors. `bge-m3` uses 1024-dimension vectors, with richer content tracking.
I had expected that switching to `bge-m3` would yield better results but that did not happen.
- **Experiment #4 vs #1: Key variable change is 2-stage retrieval using a bi-encoder followed by a cross-encoder.** Bi-encoders encode text independently for speed, while cross-encoders process text together for high accuracy. In Exp #4, I used 2-stage retrieval with a bi-encoder (model: `nomic-embed-text`) retrieving 30 chunks from the vector db, followed by a Cross-Encoder (model: `ms-marco-MiniLM-L-6-v2`) that reranked the 30 chunks to place the top 3 matches at the top.
This was expected to improve the accuracy of the pipeline but it did not.

These results pointed to the fact that the quality of data in the vector database was poor, which rendered ineffective all the techniques that I had tried. I needed a better chunking strategy.

Experiment #6: Hierarchical Chunking with Metadata

I used **parent-child chunking**, per details in the table. Short user queries match much better with small, focused child chunks than they do with larger chunks that may represent half a page of data. This boosts semantic matching and accuracy. For the matching chunks from the vector database, the parent chunks are retrieved and passed to the LLM. This ensures the LLM has the larger context for generating rich insights.

Metadata: During database ingestion, the LLM was used to extract one SWOT theme for each parent chunk, and added to the corresponding child chunks. During lookup, one or more SWOT themes are extracted from the user query using the LLM, which are then passed along with the user query. This improves accuracy by eliminating chunks with different metadata, reduces token-bias, and spurious matches with boiler-plate like statements.

These changes yielded good improvements. Token bias reduced and semantic content improved significantly. Boilerplate type statements disappeared and specificity and accuracy of the final output improved a lot.

Results

Following is the final response for Experiment #6 (Hierarchical Chunking with Metadata)

Query: What are the core market growth risks and challenges for NVDA in calendar 2026?

Final Response

Based on the provided context, I have identified the following core market growth risks and challenges to NVIDIA in calendar 2026:

1. **Export Controls:** Ongoing export control restrictions and potential expansion of these restrictions could harm NVIDIA's competitive position and impact its business, operating results, and financial condition.
2. **Licensing Process:** Delays or denials of licenses by the US Government (USG) for significant customers could lead to a loss of business opportunities and revenue.
3. **Competition from Chinese Suppliers:** The licensing process may inadvertently benefit NVIDIA's competitors in China, Europe, and Israel, who can supply semiconductor products without the same regulatory burdens.
4. **Regulatory Uncertainty:** The potential passage of legislation like RASA (Remote Access Security Act) could impose new restrictions on cloud service providers and OEMs, affecting NVIDIA's business.
5. **Customer Adoption of Alternatives:** The complexities and conditions imposed by licensing requirements may encourage customers in regions like China, the Middle East, and other emerging markets to explore alternative products from suppliers not subject to US regulations.

These market growth risks and challenges highlight the importance for NVIDIA to navigate regulatory landscapes and adapt to changing customer needs and preferences while maintaining its competitive position.